
Alzheimer's Severity Prediction

**Blood biomarkers as a
prediction of Alzheimer's
disease severity**

An unconventional model of illness



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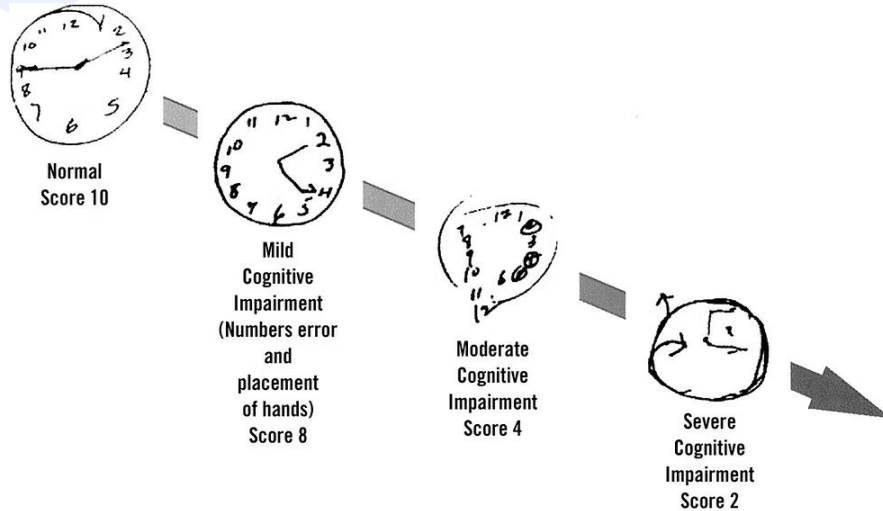
Live demo in VS code +
visualizaations

Abstract

- Background + purpose: accessibility
- The biological meaning behind our data
- Tools: LASSO and Elastic Net (biomarkers)
- Tools: random forest, and ordinal, linear, and logistic regressions
- Wrapping everything up: user interaction



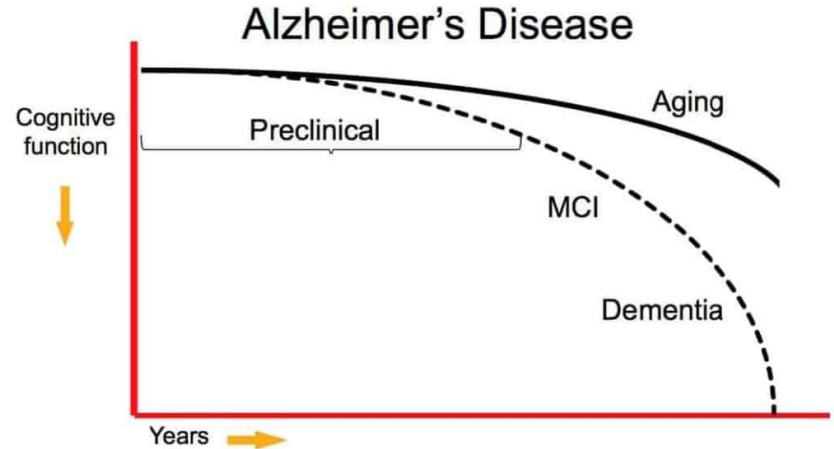
Part 1: background and purpose



Sunderland, 1989

UB

74



Accessibility is our goal

- Alzheimer's widespread disease
- Cognitive tests detect and categorize severity
 - But require qualified physician
- Leads us to our question: can blood biomarkers be used to predict Alzheimer's severity?
 - Markers of brain health obtained through blood





What are we aiming?

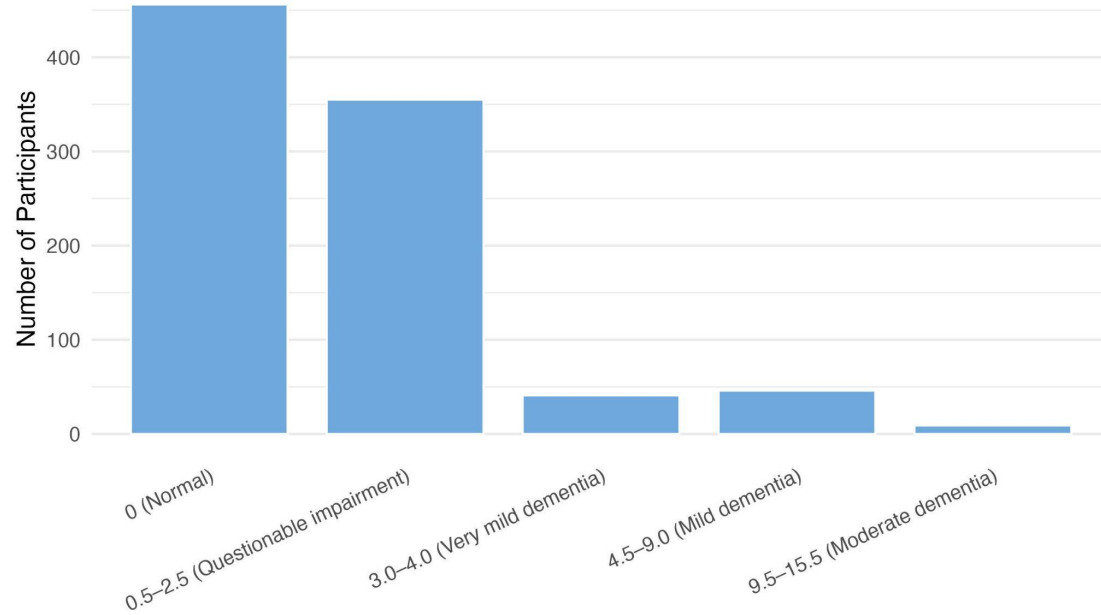
We use cognitive impairment score to link biomarker presence to disease severity.

CDR / CDR-SB

Clinical Dementia Rating

- Assess global severity
- Used CDR-SB (sum-of-boxes) for continuity
- Range 0 - 18

CDRSB Severity Distribution

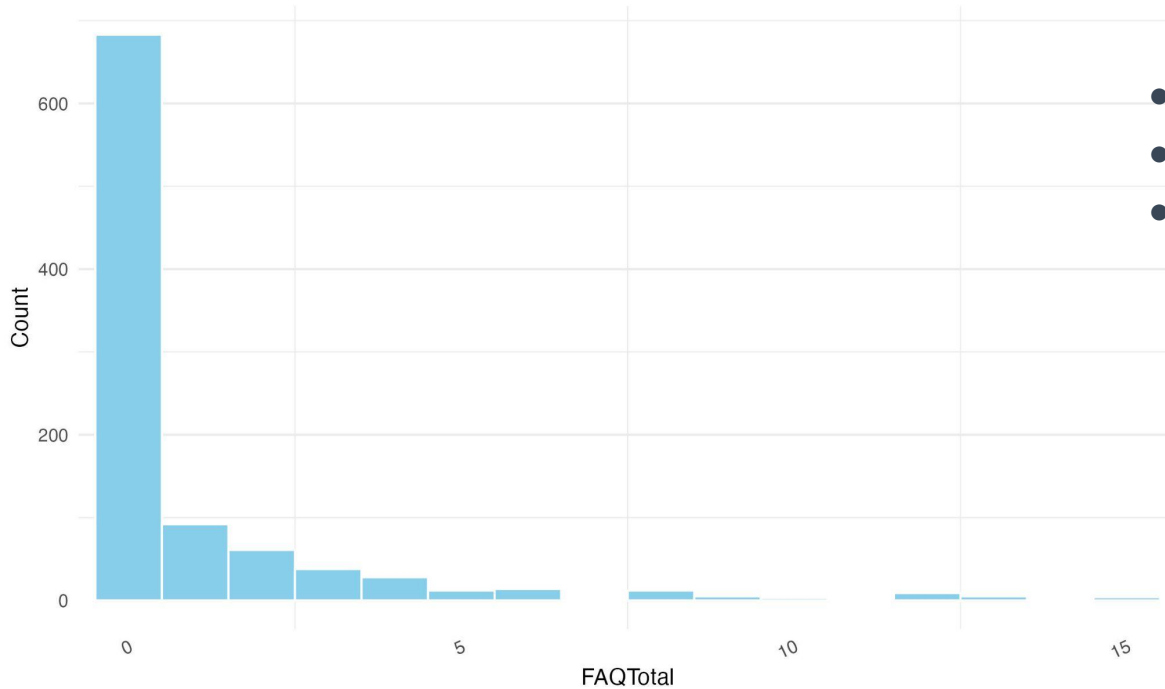


FAQ

Functional Activity Questionnaire

- Assess Functional Impairment
- Range 0 – 30
- Higher Score -> higher impairment

Distribution of FAQ



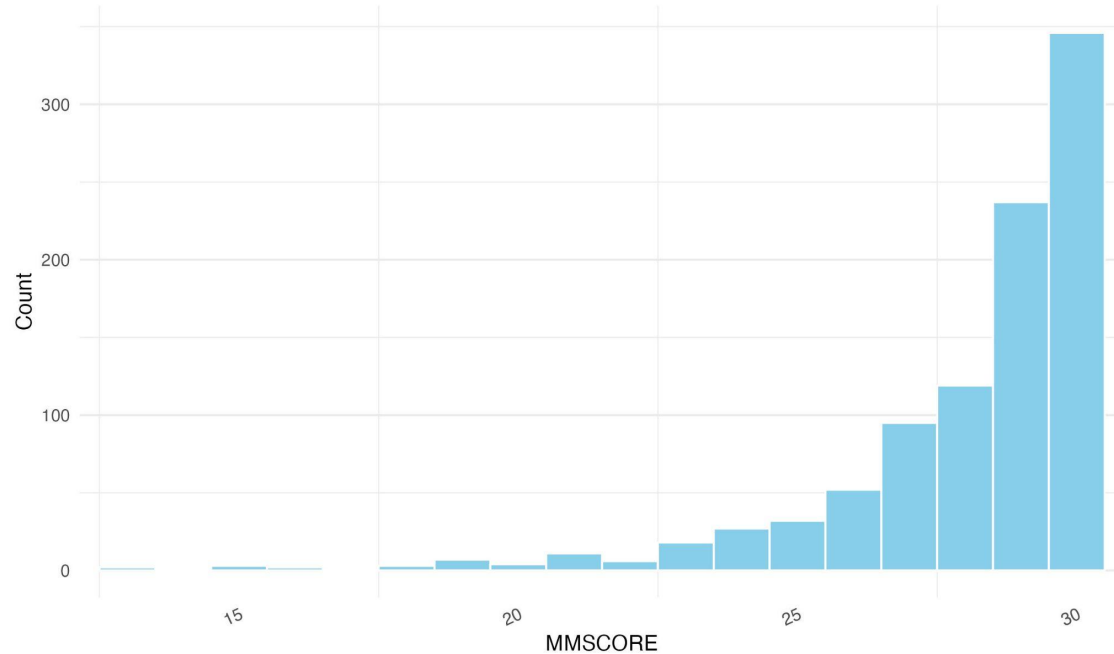
MMSE

Mini-Mental State Examination

- Assess Cognitive Performance
- Range 0 - 30
- Lower score -> Higher impairment



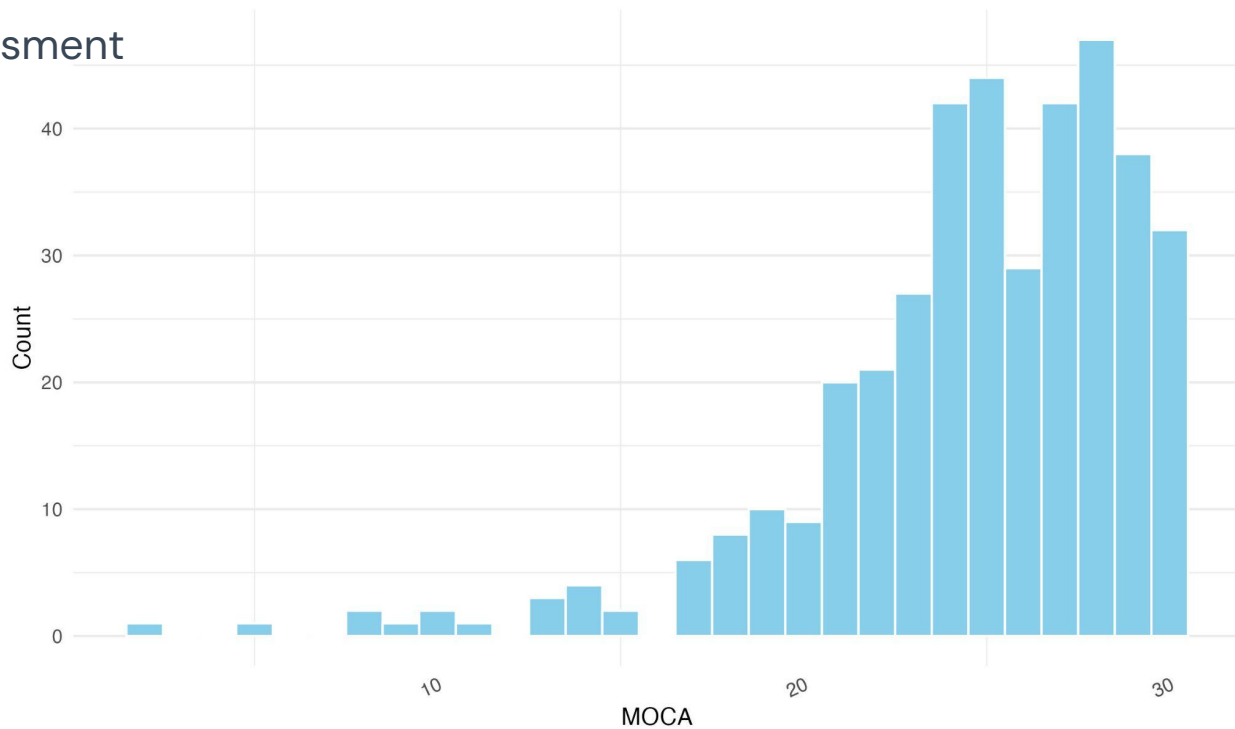
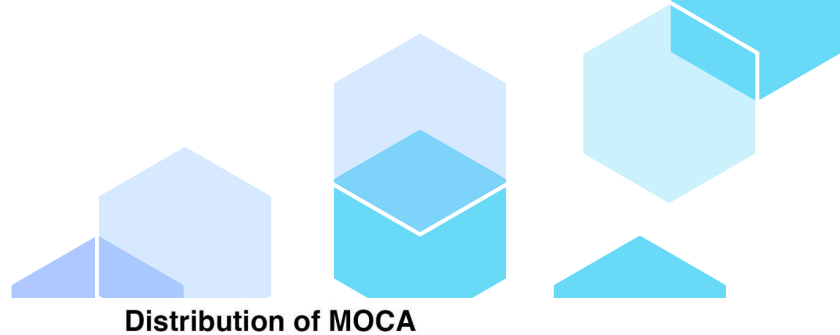
Distribution of MMSE



MOCA

Montreal Cognitive Assessment

- Assess Cognitive Performance
- Range 0 - 30

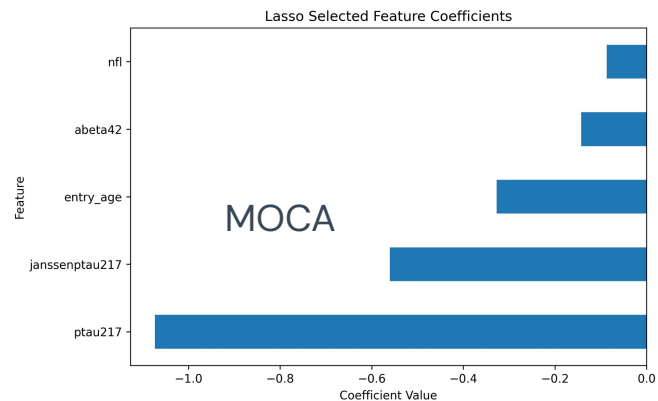
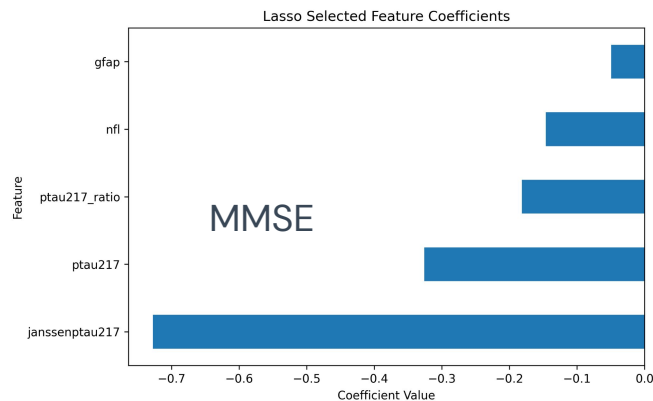
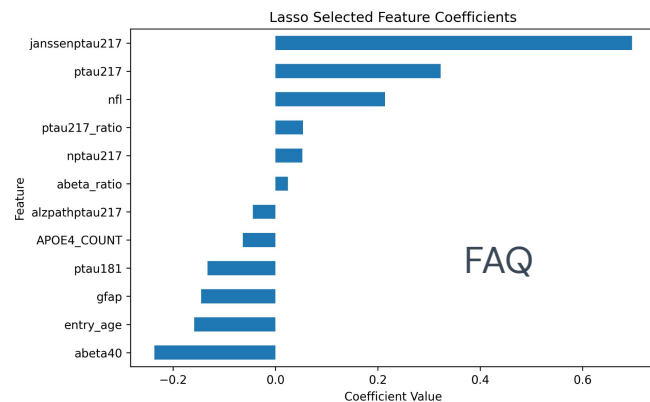
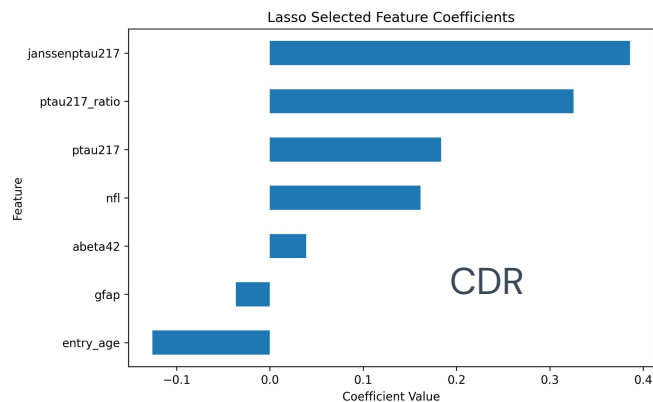


Part 2: Feature selection

- Model used: Lasso regression & Elastic Net
- Using these two models we determined the features we want to use to assess cognitive impair score by looking at biomarkers of the 4 datasets
- Features we ended up using:
 - Janssenptau217
 - Nfl
 - entry_age
 - gfap
 - Abeta42
 - APOE4_COUNT



Lasso Regression



Elastic Net

To identify high-impact blood-based biomarkers across different datasets

Feature	Coefficient
janssenptau217	0.361462
ptau217_ratio	0.319432
ptau217	0.229454
nfl	0.195737
entry_age	-0.138926
gfap	-0.070455
abeta42	0.056117
nptau217	-0.021121
ptau181	-0.016936
APOE4_COUNT	-0.004265
abeta40	0.000000
abeta_ratio	0.000000
alzpathptau217	0.000000

CDR

Feature	Coefficient
janssenptau217	0.561174
ptau217	0.292047
abeta40	-0.205998
nfl	0.172726
entry_age	-0.156120
ptau217_ratio	0.149553
gfap	-0.129308
nptau217	0.092169
ptau181	-0.091604
APOE4_COUNT	-0.053210
abeta_ratio	0.024574
abeta42	0.000000
alzpathptau217	-0.000000

FAQ

Feature	Coefficient
janssenptau217	-0.614280
ptau217	-0.324422
ptau217_ratio	-0.238940
nfl	-0.153204
gfap	-0.068487
entry_age	-0.002007
abeta40	0.000000
abeta42	-0.000000
abeta_ratio	-0.000000
alzpathptau217	-0.000000
nptau217	-0.000000
ptau181	-0.000000
APOE4_COUNT	-0.000000

MMSE

Feature	Coefficient
ptau217	-0.731372
janssenptau217	-0.684918
entry_age	-0.362538
ptau217_ratio	-0.349431
abeta42	-0.264212
nfl	-0.205274
APOE4_COUNT	0.122760
abeta_ratio	-0.007696
abeta40	-0.000000
alzpathptau217	-0.000000
gfap	-0.000000
nptau217	0.000000
ptau181	0.000000

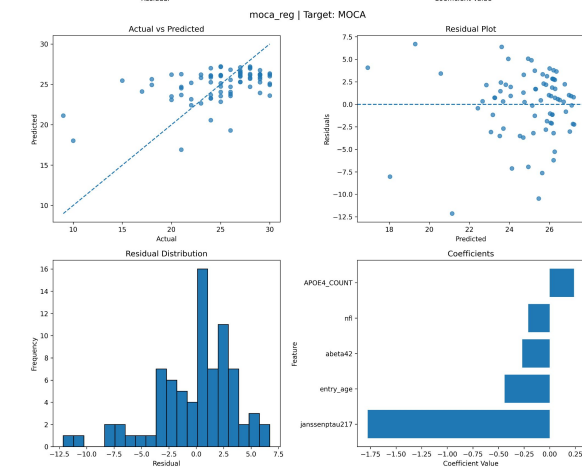
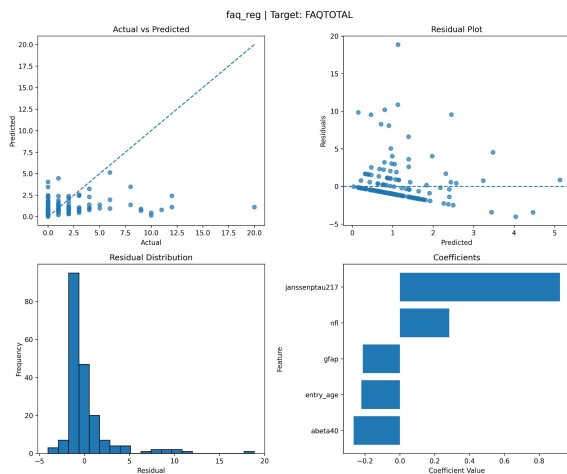
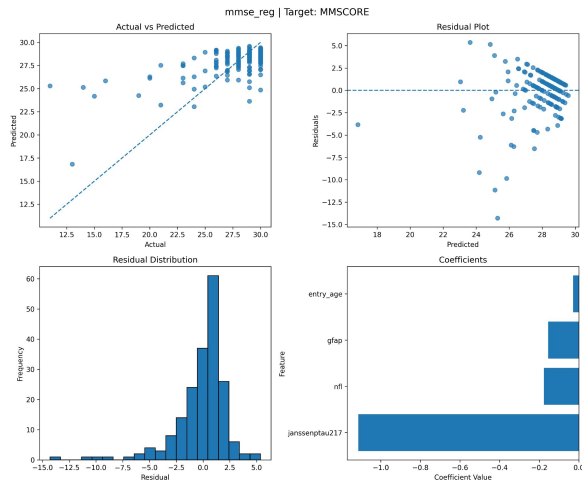
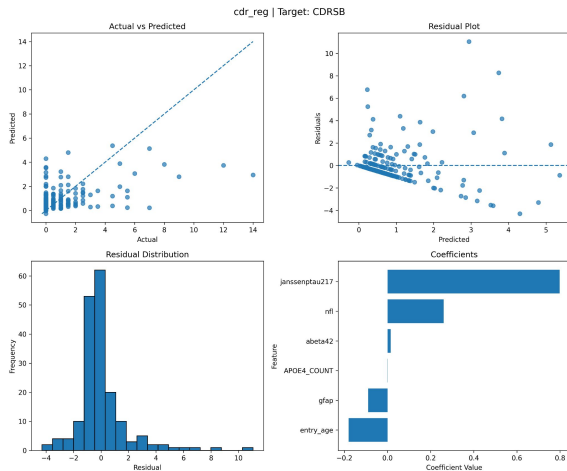
MOCA



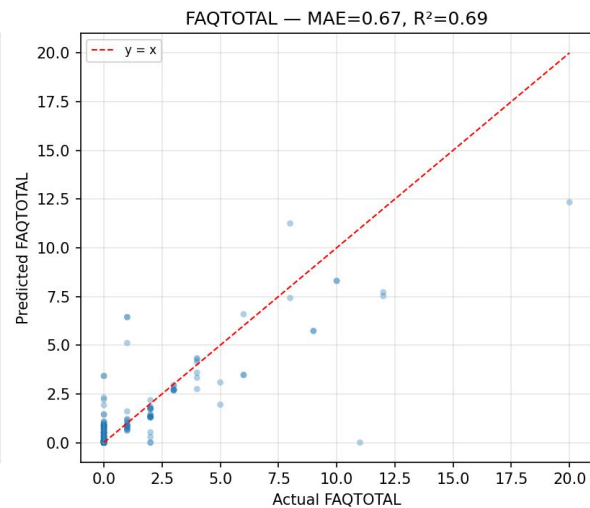
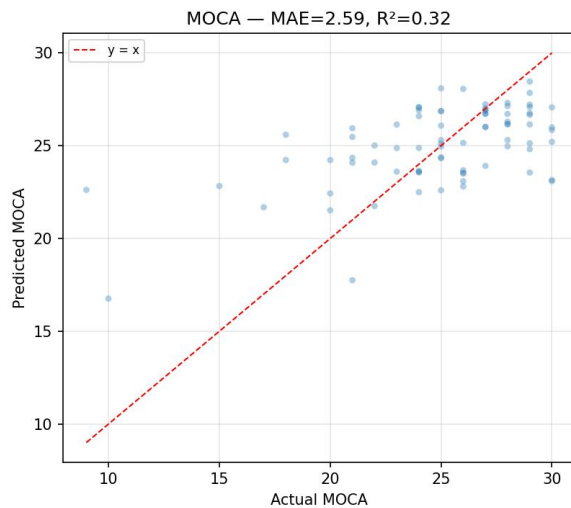
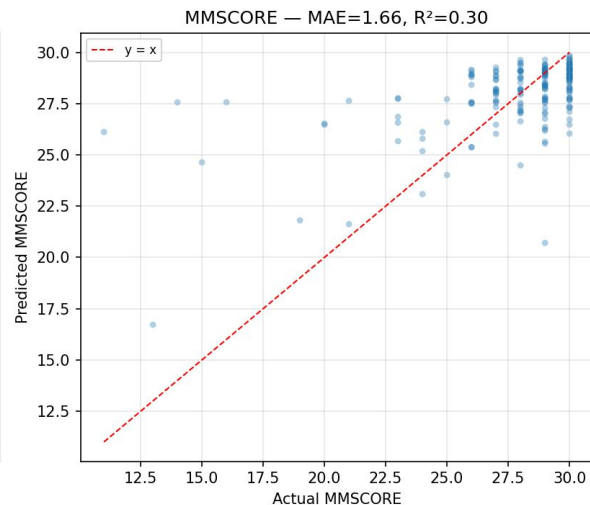
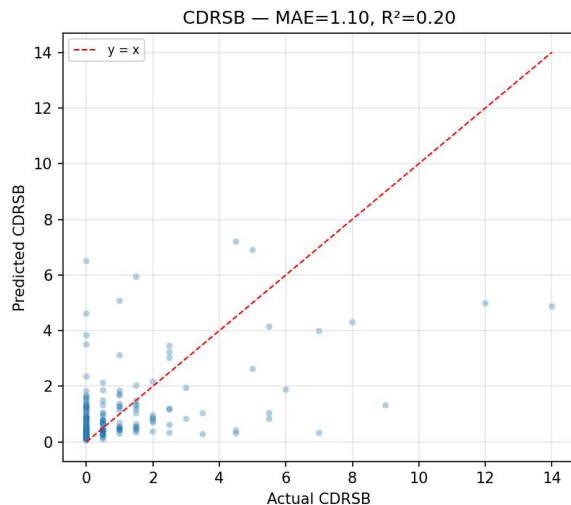
Part 3: Predict Cog test Score

- Model used: Linear regression & Random forest
- The results indicated that the random forest model is a better predictor of CDR, FAQ, and MOCA. Linear Regression works better on MMSE.

Linear Regression For CDR/FAQ/MMSE/MOCA

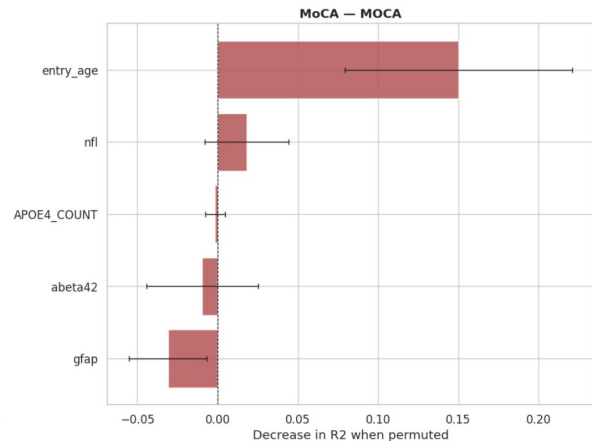
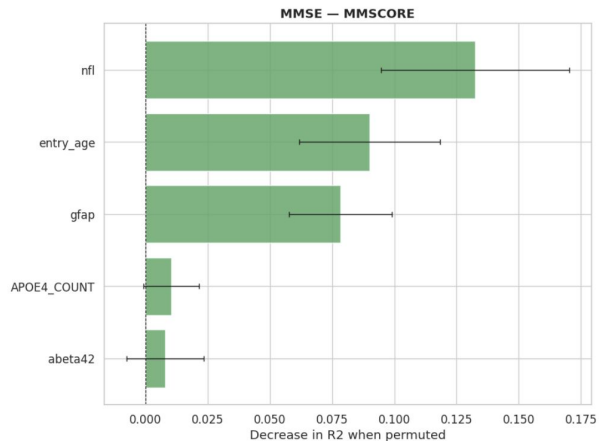
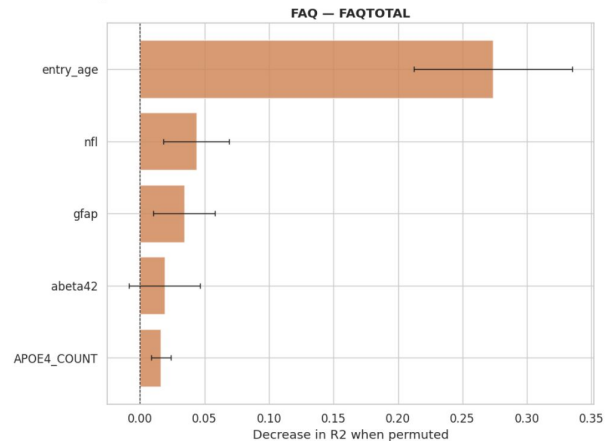
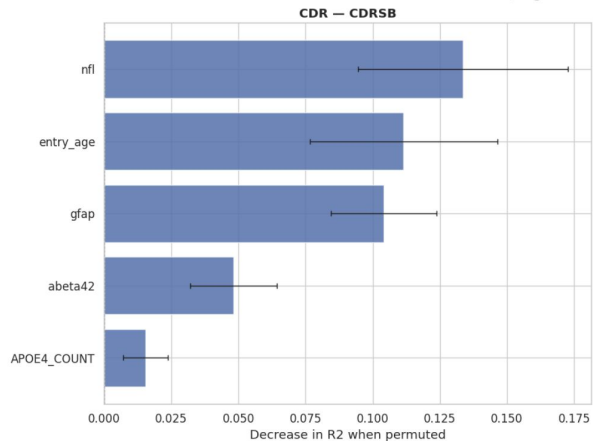


Random Forest regressor: actual vs predicted (held-out test)



Effect of Permutations on R^2

Permutation importances — test set
(negative = noise feature)



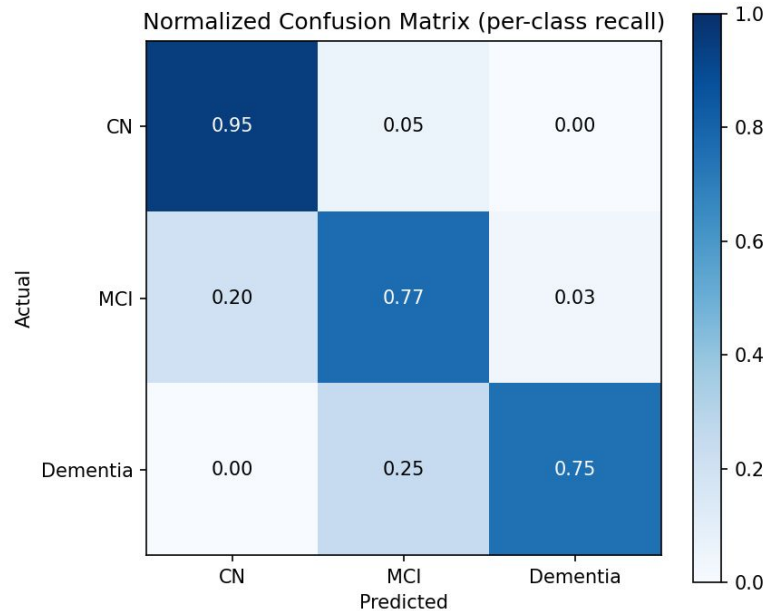
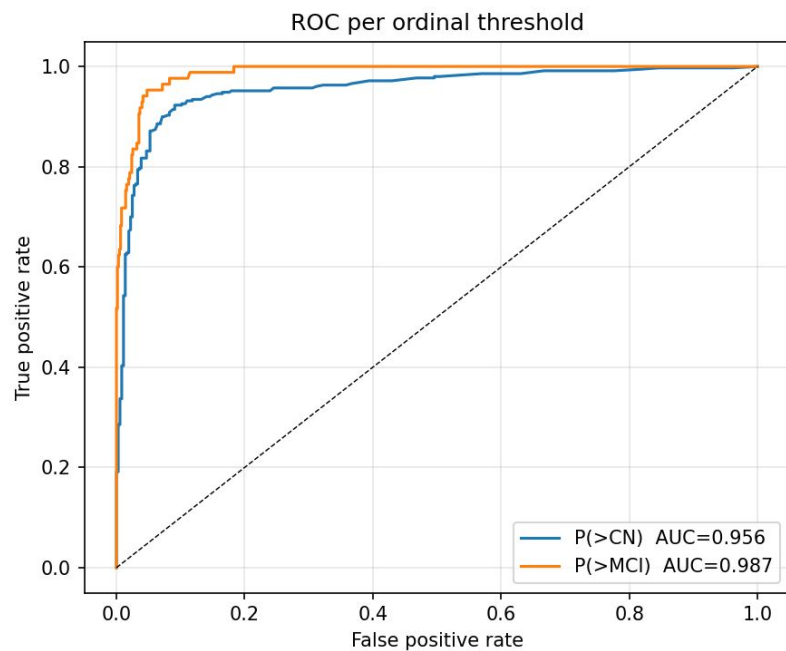


Part 4: Cognitive score to disease diagnosis

Aim: Severity Diagnosis:

- CN
No cognitive impairment detected.
- MCI
Mild Cognitive Impairment — monitor closely, not yet dementia.
- Dementia
Dementia-level impairment — clinical follow-up strongly advised.

Cognitive Score → Disease Diagnosis



Part 5: Final Performance

- 72.8% accuracy with no/little crucial error.
- Asking the user for 4 biomarkers, age, and APOE4, which is generally affordable and accessible.
- Produce a detailed report with predicted cognitive test score, diagnosis and suggestions.



Sample Run:

Asking for input ↓

Detailed Report generated ↓

```
=====
Alzheimer's Severity Prediction Pipeline
=====

[1/3] Loading trained pipeline from /Users/zoeyd/Desktop/UBIC_Ha
[2/3] Collecting user input...

Enter biomarker values (press Enter to use training median):
APOE4_COUNT: 0
abeta42: 61.516
entry_age: 72.65
gfap: 265
janssenptau217: 0.016
nfl: 31.939

[3/3] Generating predictions...

=====
PATIENT SUMMARY
=====

Biomarker inputs used:
APOE4_COUNT      0.0      normal: 0 - 1 alleles
abeta42          61.516   normal: 29.17 - 54.78 pg/mL
entry_age        72.65
gfap             265.0     normal: 65.8 - 355.8 pg/mL (for age 73)
janssenptau217   0.016   normal: 0.023 - 0.145 pg/mL
nfl              31.939   normal: 2.81 - 40.61 pg/mL

Predicted clinical scores:
CDR Sum of Boxes      0.23 (scale 0-18, higher = worse)
Mini-Mental State Exam 27.23 (scale 0-30, lower = worse)
Montreal Cog Assessment 22.83 (scale 0-30, lower = worse)
Functional Activities Q 0.00 (scale 0-30, higher = worse)

Severity classification: CN (confidence 70.4%)

Probability per class:
CN      0.704 ██████████
Dementia 0.002
MCI      0.294 ████████

Interpretation:
>>> CN      No cognitive impairment detected.
MCI      Mild Cognitive Impairment - monitor closely, not yet dementia.
Dementia  Dementia-level impairment - clinical follow-up strongly advised.

5 most similar patients in reference dataset:
RID      DATE      CDRSB  MMSCORE  MOCA  FAQTOTAL  DIAGNOSIS  distance
4855 2018-02-09  0.0    27.0    23.0    0.0      CN  0.367015
6346 2018-05-02  0.0    27.0    23.0    0.0      CN  0.367015
6391 2018-05-23  0.0    27.0    23.0    0.0      CN  0.367015
6442 2018-06-15  0.0    27.0    23.0    0.0      CN  0.367015
6624 2018-10-30  0.0    27.0    23.0    0.0      CN  0.367015

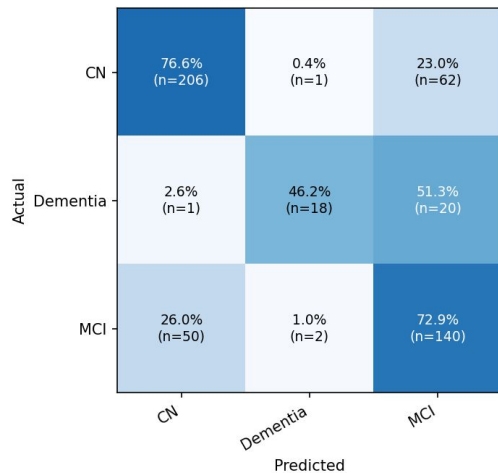
Note: Research tool only. Not a clinical diagnosis.
=====
```

Real Patient Data:

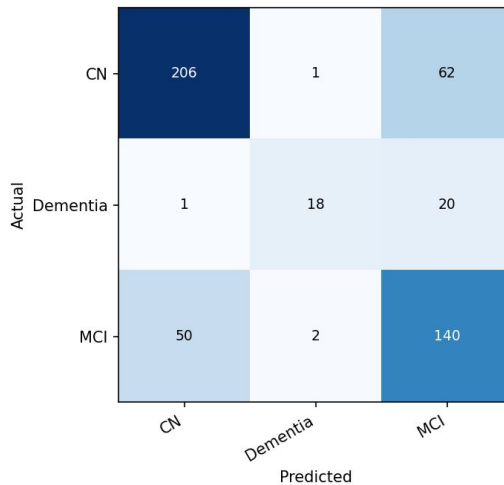
APOE4_COUNT: 0 CDRSB: 0.0
abeta42: 61.516 MMSE: 27.0
entry_age: 72.65 MoCA: N/A (missing)
gfap: 265.0 FAQ: 0.0
janssenptau217: 0.016 Diagnosis: CN
nfl: 31.939

What's Good/bad about our model?

Biomarkers → Diagnosis (end-to-end, row-normalized)
acc=72.8% n=500



Biomarkers → Diagnosis (end-to-end)
acc=72.8% n=500

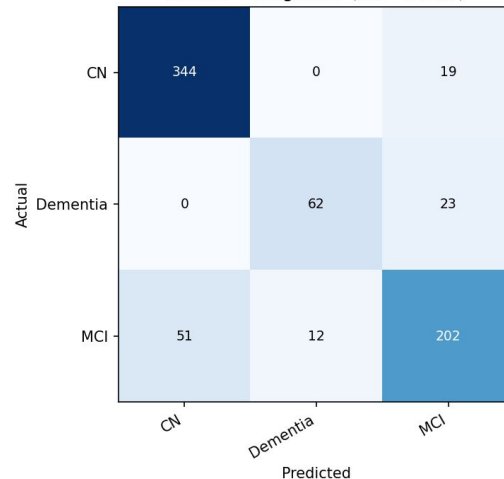


Weakness: Sometimes missed Dementia as MCI

Strength:

- 72.8% accuracy on 500 samples from biomarkers alone
- Good separation of CN and MCI
- Very few extreme errors between CN and Dementia
- Most mistakes are between adjacent stages, which is clinically reasonable

Scores → Diagnosis (acc=85.3%)



Challenge we encountered:

- Biomarkers are not naturally directly linked to Cognitive tests
- Not enough Data (already filtered the entire ADNI dataset)
- DataHub!





Thank you!

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